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Studierichting: Informatica

Oefeningenreeks 4A nr. 45.1

$$\begin{aligned}\int \frac{9x^2 - x + 2}{4x^5 + x^3} dx &= \int \frac{9x^2 - x + 2}{x^3(4x^2 + 1)} dx = \frac{1}{4} \int \frac{9x^2 - x + 2}{x^3 \left(x^2 + \frac{1}{4}\right)} dx \\&\rightarrow \frac{9x^2 - x + 2}{x^3 \left(x^2 + \frac{1}{4}\right)} = \frac{A}{x^3} + \frac{B}{x^2} + \frac{C}{x} + \frac{Dx + E}{x^2 + \frac{1}{4}} \\&\Rightarrow A = \frac{2}{\frac{1}{4}} = 8 \\&\rightarrow \frac{9x^2 - x + 2}{x^3 \left(x^2 + \frac{1}{4}\right)} - \frac{8}{x^3} = \frac{x - 1}{x^2 \left(x^2 + \frac{1}{4}\right)} \\&\Rightarrow B = \frac{-1}{\frac{1}{4}} = -4 \\&\rightarrow \frac{x - 1}{x^2 \left(x^2 + \frac{1}{4}\right)} - \frac{-4}{x^2} = \frac{4x + 1}{x \left(x^2 + \frac{1}{4}\right)} \\&\Rightarrow C = \frac{1}{\frac{1}{4}} = 4 \\&\Rightarrow D \left(\frac{1}{2}i\right) + E = \frac{-\frac{9}{4} - \frac{1}{2}i + 2}{-\frac{1}{4} \cdot \frac{1}{2}i} = 4 - 2i \Rightarrow E = 4 \\&\Rightarrow \frac{1}{2}Di = -2i \Rightarrow D = -4 \\&= 2 \int \frac{1}{x^3} dx - \int \frac{1}{x^2} dx + \int \frac{1}{x} dx + \frac{1}{4} \int \frac{-4x + 4}{x^2 + \frac{1}{4}} dx \\&= -\frac{1}{x^2} + \frac{1}{x} + \ln|x| + \frac{1}{4} \int \frac{-2(2x) + 4}{x^2 + \frac{1}{4}} dx \\&= -\frac{1}{x^2} + \frac{1}{x} + \ln|x| - \frac{1}{2} \ln \left|x^2 + \frac{1}{4}\right| + \frac{1}{\frac{1}{2}} \text{Bgtan} \left(\frac{1}{\frac{1}{2}}x\right) + c \\&= -\frac{1}{x^2} + \frac{1}{x} + \ln|x| - \frac{1}{2} \ln \left|x^2 + \frac{1}{4}\right| + 2 \text{Bgtan}(2x) + c\end{aligned}$$

References